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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS403 - MICROPROCESSOR AND EMBEDDED SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** Questions from **Unit – I & Unit – II** each and **One full** Question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the different types of Addressing modes of Intel 8086 Microprocessor. Explain with the help of example for each.	10	L*2	1	1
	b) Develop an assembly language program to search an 8-bit unsigned number in an array using binary search algorithm. Write appropriate comments.	10	L3	1	3
2.	a) Explain the various flag registers used in Intel 8086 microprocessor. Explain each in detail.	10	L2	1	1
	b) Develop an assembly language program to check if two strings are equal or not. Assume both strings are available in the data segment.	10	L3	1	1
3.	a) Give the interrupts and the necessary parameters to perform the following operations in Intel 8086 microprocessor: i. Get the cursor position ii. Get the system date iii. Create a File iv. Delete a sub directory	10	L2	2	1
	b) Explain the following 8086 microprocessor instructions by giving an example for each: i. DAA ii. AAM iii. Shift instructions iv. Rotate instructions	10	L3	2	2
Unit – II					
4.	a) Explain the embedded system software in detail.	10	L2	2	2
	b) Develop an assembly language procedure to convert 8-bit BCD to equivalent Hexadecimal values.	10	L3	2	2
5.	a) Briefly explain the RISC design philosophy. Bring out the differences between RISC and CISC machines.	10	L2	4	1
	b) Write a embedded C program to drive a stepper motor clockwise and anticlockwise with the help of switches.	10	L2	4	1
6.	a) With a neat block diagram explain the ARM core dataflow model.	10	L2	3	1
	b) Write an embedded C program to perform ring counter on a 7-segment display.	10	L2	3	1
Unit – III					
7.	a) With a neat block diagram explain the working of an 8259A Priority Interrupt Controller.	10	L2	5	1

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| b) | Explain the minimum mode configuration of 8086 in detail. | 10 | L2 | 5 | 1 |
| 8. a) | Explain the following interrupt types in detail: | | | | |
| | i. Type 0 | | | | |
| | ii. Type 1 | | | | |
| | iii. Type 2 | 10 | L2 | 5 | 1 |
| b) | Convert the following instructions into machine language form: | | | | |
| | i. MOV 43H[SI], DH | | | | |
| | ii. MOV CX, [437AH] | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
